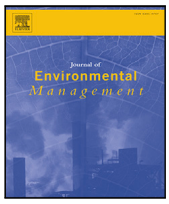
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Research article



Synthesising textual- and model-based evidence for proactive urban policymaking: A case in air quality management

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ABSTRACT

Urban governance is hindered by the disconnect between qualitative knowledge embedded in textual sources and the predictive capabilities of quantitative models, making proactive policymaking challenging. This study proposes and validates a two-phase framework that automates the 'dialogue' between the evidence paradigms for urban air quality management. In the first phase, a Retrieval-Augmented Generation (RAG) pipeline synthesises extensive literature into a structured knowledge graph, generating evidence-based policy hypotheses. This approach navigates the challenge of evaluating overwhelming potential policy combinations. Following this, these qualitative hypotheses are automatically translated into simulation scenarios and subjected to rigorous, ex-ante quantitative testing, providing the rapid feedback necessary to overcome slow policy evaluation cycles. A case study in London, addressing post-2023 ULEZ expansion air quality challenges, demonstrates the framework's real-world utility. The RAG identified a portfolio of next-generation interventions targeting heating systems, construction machinery, and solvent emissions. Subsequent simulation with the SHERPA model produced spatial impact maps of pollution-level reductions and, critically, uncovered complex system dynamics. The framework's ability to efficiently screen a broad portfolio of options while revealing such complex consequences underscores its value for proactive governance. By integrating textual and simulation evidence into a scalable methodology, this research provides a robust tool to enable more effective, evidence-informed urban environmental policy.

1. Introduction

Rapid urbanisation creates complex socio-ecological systems where dense populations with diverse and conflicting needs strain urban environments and infrastructure (Schindler et al., 2018). This complexity presents a significant challenge for urban governance, where the conventional approaches are typically fragmented across agencies and mostly reactive to crises. A paradigm shift toward more proactive, evidence-informed governance is underway (Head, 2016), redefining public policies as key instruments for translating evidence and foresight into action (Hossin et al., 2023). This shift, however, hinges on bridging the well-documented gap between knowledge and action.

In urban environments, this gap persists because establishing a clear understanding of the causal relationship between a policy and its effects is often challenging (Schlüter et al., 2023). An individual evaluation might show that a policy intervention worked in a specific context, but it could fail to explain the underlying causal mechanisms, i.e., the 'how' and 'why'. This challenge is not merely technical: it results from policies intersecting with a dense web of real-world constraints. These include institutional barriers like fragmented governance, political pressures from vested interests, and crucial socio-economic and equity dimensions, which are often invisible to conventional analysis (Freund et al., 2019). The root of this oversight lies in a foundational '*Qualitative-Quantitative Divide*' that pervades environmental science. Dominant analytical tools are designed to process structured, numerical data but are structurally blind to the rich qualitative information found in legal texts, stakeholder testimonies, and community feedback, where these real-world constraints and human knowledge are described. In this study, evidence synthesis is defined as the systematic process of bridging this divide: gathering and integrating both quantitative and qualitative knowledge to answer specific questions, providing urban governance with a rigorous means of clarifying causal chains and transforming fragmented evidence into reliable, policy-relevant knowledge for policymaking in complex socio-political contexts (Cooke et al., 2023).

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To navigate the causal relationships within such complex systems, two dominant approaches are used to generate policy-relevant evidence: empirical analysis of real-world observations and computational modelling of simulated systems (Schlüter et al., 2023). The empirical approach evaluates causal relationships and identifies causal mechanisms by analysing quantitative or qualitative data from real-world observations (Kiekens et al., 2022). On the other hand, the modelling approach uses process- or agent-based simulations to investigate the effects of changing system variables and to identify causal mechanisms under controlled, simulated conditions (Assaf et al., 2023).

The challenges of evidence synthesis to support proactive policymaking are manifest in the context of 'wicked' urban problems, where evidence is vast, fragmented, and contested. The domain of urban air quality management perfectly illustrates these challenges, making it a critical case study (Amann et al., 2020). Conventional air quality policymaking faces both structural and temporal constraints that hinder the synthesis of causal evidence. Structurally, fragmented governance, marked by jurisdictional ambiguity and siloed data across agencies, leads to uncoordinated or conflicting mitigation efforts (Sofia et al., 2020). For instance, the report by the National Audit Office found that the central NO_2-compliance programme in London has suffered delays and unclear oversight owing to mismatches between national road-network management and local air quality actions.[1] This is compounded by the temporal challenge of slow feedback loops, where full consequences of policy decisions may not be observable for years. For example, the Ultra Low Emission Zone (ULEZ) in London produced only modest improvement in air quality in its first year, but more substantial reductions became clear after several years of phased expansions.[2] This time lag makes it difficult to reliably establish causal evidence between a policy intervention and its outcomes based on real-world observations (Feng et al., 2023). Another significant barrier to effective, evidence-informed policymaking in this domain is combinatorial complexity. The multitude of interconnected emission sources and socio-economic factors creates a vast solution space. With dozens of potential policy levers, a recent review identified 58 distinct traffic interventions that combine into over a thousand scenarios, illustrating how the number of possible strategies grows exponentially (Khreis et al., 2023). This 'combinatorial explosion' renders existing evidence synthesis strategies, which are designed to evaluate a small number of pre-selected policies rigorously, computationally and practically infeasible for the crucial task of policy screening. A critical gap in the early stages of policymaking is the lack of a mechanism for the broad, rapid screening of numerous potential policy interventions.

Conventional evidence synthesis methods, while valuable for in-depth evaluation, fail for policy screening due to the challenge of combinatorial complexity. This study proposes and demonstrates a framework explicitly designed to address this bottleneck, enabling a more proactive and scalable approach to identifying promising interventions within a vast policy landscape. By coupling Large Language Model (LLM) guided evidence synthesis with model-based simulation, this study showcases a more transparent and evidence-based approach that automates the 'dialogue' between qualitative evidence and quantitative simulation, highlighting a significant step forward in urban air quality policymaking.

The remainder of the paper is organised as follows: Section 2 reviews the paradigms of evidence synthesis, i.e., the empirical and modelling approaches, and their application in air quality policymaking, establishing the need for an integrated 'dialogue' framework. Section 3 introduces the proposed two-phase framework that operationalises this dialogue, detailing the knowledge-based synthesis for hypothesis generation and the simulation-based hypothesis testing. Section 4 demonstrates the framework's application through a case study in London. Section 5 discusses the broader implications of the framework and proposes a vision for a more comprehensive evidence cycle. Finally, Section 6 summarises key findings.

[1] https://www.nao.org.uk/reports/tackling-local-breaches-of-air-quality/.

[2] https://content.tfl.gov.uk/london-wide-ulez-one-year-report.pdf.

2. Literature review

2.1. Empirical analysis and computational modelling for evidence synthesis

The synthesis of causal evidence in complex urban systems is dominated by two approaches: the empirical analysis of real-world observations and the computational modelling of simulated systems, as described in Table 1. Each offers distinct approaches for generating policy-relevant knowledge, but is constrained by inherent limitations.

The empirical approach evaluates causal relationships by analysing data from real-world observations. It offers high external validity, as its conclusions are grounded in real-world conditions, but suffers from low internal validity because isolating the policies' causal effect from confounding variables in uncontrolled environments is difficult. This approach can be broadly divided into qualitative and quantitative methods (Houghton and Paniagua-Avila, 2023; Kiekens et al., 2022).

Qualitative analysis involves the in-depth examination of case studies, policy documents, and reports to understand causal processes. For example, process tracing is used to trace the pathway between a cause and an effect in a particular case, providing a rich and contextualised understanding of how outcomes occurred (Carodenuto et al., 2024). However, manual analysis of vast text repositories is labour-intensive and subjective (Hooper et al., 2024). Latest research has explored the Natural Language Processing (NLP) or LLMs to improve the efficiency of processing text data (Hong et al., 2025; Nie and Liu, 2025). While LLMs demonstrate emergent capabilities to synthesise policy-relevant insights, their integration into public governance is constrained by fundamental structural and epistemic limitations. A primary concern involves misinformation and hallucinations. As these LLMs rely on probabilistic pattern matching rather than factual accuracy, they frequently generate ungrounded content that can undermine the epistemic basis of democratic policymaking (Shah et al., 2024). This risk is exacerbated by the algorithmic bias. LLMs exhibit significant disparities across gender, ethnicity, and social class (Coeckelbergh, 2025). In environmental governance, this risks 'automating inequality' by entrenching historical disparities in data availability and favouring dominant social perspectives at the expense of marginalised communities. While Chain-of-Thought (CoT) prompting can enhance logical consistency through structured task design and reasoning optimisation, it does not guarantee 'faithfulness', often producing plausible but misleading steps that mask internal decision drivers or skip the 'interpretive labour' essential for sound professional judgment (Lu et al., 2022). Recognising these vulnerabilities, recent scholarship suggests that LLMs must be deployed as regulated decision-making infrastructure rather than autonomous agents. This necessitates architectures such as Retrieval-Augmented Generation (RAG), which ground model reasoning in the structured, validated facts of a Knowledge Graph, serving as a necessary 'epistemic anchor' (Arazzi et al., 2025).

Quantitative analysis employs machine learning or statistical techniques on vast, real-time datasets from sensor networks to evaluate policy impacts and untangle complex causal dynamics between interventions and outcomes, including the cascade effects within the urban environment (Fu et al., 2025). However, their effectiveness is often constrained by issues of data quality and spatial coverage, as sparse sensor networks can provide a fragmented and incomplete picture of a highly localised phenomenon, undermining the reliability of the analysis (Qu et al., 2023).

In contrast, the modelling approach uses simulations to investigate causal mechanisms under controlled conditions (Assaf et al., 2023). This approach employs various models: process-based models formalise physical and chemical processes to anticipate intervention impacts, and agent-based models simulate individual actor interactions between individual actors within a socio-ecological context to understand the collective behaviours and system-level patterns emerging from micro-level decisions. While models enable forward-looking, scenario-based analysis that is impossible in the real world, the external validity of this

Table 1
Comparison of empirical and modelling approaches for evidence synthesis.

	Empirical approaches	Modelling approaches
Description	Analyse data from real-world observations, using qualitative methods to extract insights from vast textual data (e.g., policy documents) and quantitative methods that use machine learning to learn patterns from sensor data.	Use simulations to investigate causal mechanisms under controlled conditions, enabling forward-looking, scenario-based analysis of policy interventions that is impossible in the real world.
Research design	Research designs include in-depth qualitative analysis of single or multiple case studies and quantitative experiments using real-time sensor data.	Research designs involve applying existing process-based models to simulate scenarios with and without an intervention, or developing agent-based models that use empirical facts to study nature-society feedbacks.
Examples	Qualitative methods include process tracing and historical analysis. Quantitative methods include machine learning analysis and statistical techniques such as difference-in-differences and matching.	Methods include mathematical formalisations of environmental processes in process-based models, and numerical or agent-based simulations of interactions between agents in a social-ecological context.

approach is always conditional. Any causal claims derived from a model are entirely contingent on the realism of its underlying assumptions and the quality of its input data (Banitz et al., 2022). This is compounded by another key practical limitation: the computational inefficiency of running high-resolution simulations for every possible scenario often renders the approach infeasible within policy cycles.

Given the complementary strengths, a growing body of literature argues that robust causal understanding requires a deeper integration of empirical and modelling approaches. Integrating the real-world grounding of empirical analysis with the controlled causal-testing power of simulation yields several benefits: it can improve the design of empirical studies, reduce uncertainty in causal estimates, and elucidate complex dynamics that neither approach could uncover in isolation. As illustrated in Fig. 1, the most straightforward form of integration is through comparison, where independent empirical and modelling studies addressing the same causal question are conducted to triangulate findings. For example, studies have compared the ex-ante predictions of energy policy impacts from models with the ex-post results from empirical evaluations to strengthen the causal claim (Qiu et al., 2020). A more involved strategy is via complementarity, where one approach is used to expand on the insights of the other. For instance, an empirical study measured the observed impacts of a conservation program in Costa Rica. A biophysical model complemented the analysis by estimating the unobservable counterfactual: 'what would have happened to local poverty and ecosystem services in the program's absence', an insight the empirical data alone could not provide (Ferraro et al., 2015). The most interactive of these strategies is by dialogue, which can occur in two ways. In the first kind, a modelling study is conducted to inform and improve the design of an empirical study. For example, an air quality model can simulate how pollutants are transported by wind, allowing researchers to design a policy experiment with spatial clusters large enough to capture these 'spillover' effects (Schlüter et al., 2023). In the second kind, an empirical study informs the design of a model. For instance, a qualitative analysis of policy documents can identify a key social mechanism, such as how interest groups build coalitions. This empirically-grounded behaviour can then be formalised as a rule in an agent-based simulation, allowing for a more realistic test of how political dynamics influence fishery governance (Orach et al., 2020).

2.2. Evidence synthesis for urban air quality management

Both the empirical and modelling approaches have real-world applications in the domain of urban air quality management for evidence synthesis. The domain serves as a critical case study, perfectly illustrating the structural, temporal, and combinatorial challenges to evidence synthesis previously outlined. The various evidence synthesis approaches and consequent policymaking adopted within air quality governance highlight their respective strengths, limitations, and the emerging need for new tools to navigate an increasingly complex policymaking landscape.

In response to the constraints of conventional, expert-led policymaking, the data-driven approach has emerged as a modern evolution of the empirical approach, generating policy-relevant evidence from real-world observations. The Internet of Things (IoT), machine learning, and big data enable real-time collection and interpretation of environmental (e.g., pollutant concentrations) and contextual (e.g., traffic flows and meteorological factors) data, enhancing the precision and responsiveness of policy assessments. Urban computing integrates IoT and cloud platforms to support predictive air quality management and emissions monitoring (Illescas-Martinez et al., 2025). Mobile sensing can reveal pollution hotspots (Zhao et al., 2021), while machine learning models simulate dispersion and evaluate interventions such as clean air zones (Farhadi et al., 2023; Molina-Gómez et al., 2021). Beyond transport, causal inference has linked clean heating in China to substantial PM2.5 reductions and health benefits (Song et al., 2023), and high-resolution inventories assist in assessing industrial retrofits and energy transitions (Gu et al., 2023). Natural experiments further demonstrate its value. COVID-19 traffic reductions lowered NO_2 in European cities but showed mixed effects on particulates and ozone, highlighting the need for multi-sectoral strategies (Volta et al., 2022). In New York, the ZAPPA framework combines emissions and health models to prioritise local interventions and support environmental justice (Shukla et al., 2022).

Despite its potential, the data-driven approach suffers from critical limitations due to sensor networks. Urban air pollution exhibits extreme spatial variability, particularly for traffic-related pollutants like NO_2, where concentrations can change dramatically over tens of metres. This variability is driven by the short atmospheric lifetime and rapid chemical reactions of pollutants in combination with local ventilation and mixing effects tied to urban morphology (e.g., street-canyons and building geometry) (Dai et al., 2022). Sparse sensor networks often fail to capture these localised hotspots, providing a fragmented and incomplete picture of community exposure (Qu et al., 2023). Furthermore, low-cost sensors are susceptible to calibration drift from environmental factors like temperature and humidity, compromising data integrity. At a system level, a lack of interoperability between different data platforms, inconsistent metadata, and issues of data ownership create institutional barriers to creating a unified monitoring system. More importantly, the data-driven approach remains confined to ex-post evaluation of a narrow set of implemented policies.

Complementing the data-driven approach, the model-informed approach represents the direct application of computational modelling to simulate pollutant dynamics. By embedding the governing physical and chemical processes, it enables forward-looking, scenario-based analyses with high internal validity, allowing causal testing beyond what is possible in real-world observation. A broad spectrum of models demonstrates its utility: Gaussian plume models such as AERMOD

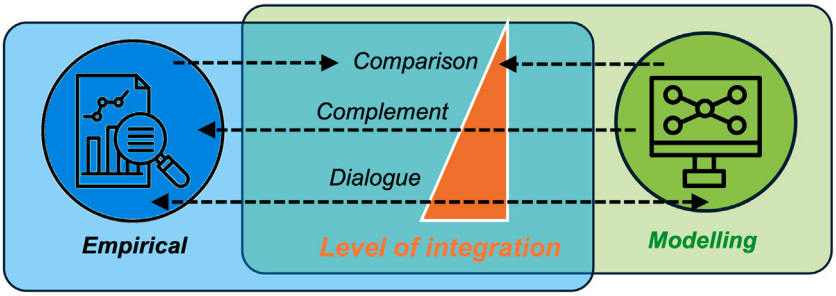


Fig. 1. Conceptual framework for integrating empirical and modelling studies.

Table 2
Comparative analysis of evidence synthesis approaches for air quality policymaking.

Approach	Advantages	Disadvantages	S	T	C
Empirical					
Conventional	Intuitive, relies on established processes.	Reactive, lacks adaptability, long policymaking cycles, slow feedback loops.	✗	✓	✗
Data-Driven	Flexible, dynamic, responsive to real-world conditions.	Evaluation is heavily reliant on sensor coverage, high investment in sensing, and data quality issues. Lack of true explainability.	✗	✗	✗
Model					
Model-Informed	Comprehensive, forward-looking, allows for optimal policy comparison under controlled conditions.	Computationally expensive, requires extensive validation, reliability is contingent on input data and assumptions.	✓	✓	✗

Note: **S** = Structural Constraints, **T** = Temporal Constraints, **C** = Combinatorial Complexity.

and ADMS-Urban are computationally efficient and widely used for regulatory and local policy assessments under constant meteorological conditions (Ridzuan et al., 2020; Mallet et al., 2018); non-steady-state models like CALPUFF simulate long-range transport and complex terrains (Holnicki et al., 2016); and fully coupled chemical transport models (e.g., WRF-Chem, CAMx) capture atmospheric chemistry and meteorology–pollution feedbacks, offering detailed, process-based insights when supported by high-resolution grids and accurate emissions inventories (Shahbazi et al., 2017). In the European context, reduced-form frameworks such as the Sharing Evidence Routine for a Person-centred plan of Action (SHERPA) model provide rapid, reproducible assessments across cities, supporting cross-comparisons and integrated planning under demanding policy cycles (Thunis et al., 2016).

The reliability of models, however, is heavily dependent on the quality and completeness of the input data, such as emission inventories and meteorological observations, which are often inconsistent or subject to significant uncertainty. Any causal claims derived from a model are entirely contingent on the realism of its underlying assumptions, compromising the direct applicability of its findings to the real world. The computational inefficiency brings further challenges. Simulating every possible policy scenario with high-resolution models is often infeasible within the time frames of urban policy cycles. This computational bottleneck hinders the use of models for the kind of rapid, iterative decision-making required for adaptive governance, unless supported by simplification techniques or pre-screening scenarios. The comparison of different approaches is illustrated in Table 2.

These limitations are not independent but are critically intertwined. The ex-post nature of the data-driven approach creates the demand for the ex-ante predictive power of simulation. Simultaneously, the combinatorial complexity inherent in wicked problems renders a purely model-informed approach computationally infeasible for broad policy screening. This creates a fundamental methodological impasse: the approach grounded in reality cannot look forward easily, and the approach that can look forward cannot feasibly navigate the vast landscape of policy options. To fill this gap, a novel framework is proposed that automates a formal 'dialogue' between the vast, unstructured knowledge in the policy literature and the rigorous testing power of computational models. Instead of treating textual evidence as mere background context, the texts are treated as the foundation of quantitative analysis. This approach systematically extracts high-level causal knowledge on policy mechanisms, barriers, and dynamics, and uses it to guide and constrain simulation models. This results in a capacity for the rapid, evidence-grounded screening of numerous policy options sand quantitatively test their potential impacts, directly confronting the complexity that has long stalled proactive environmental management.

3. Methodology

This study proposes a two-phase framework that operationalises the 'dialogue' between qualitative evidence and quantitative simulation, creating a more proactive and robust pathway for evidence-informed policymaking in urban air quality management. The design of this framework, elaborated in Fig. 2, is a direct response to a fundamental methodological impasse created by the challenges of 'combinatorial complexity' and 'slow feedback loops'. The process begins with ① Knowledge Extraction, where unstructured textual inputs are synthesised into a structured Knowledge Graph. This supports ② Hypothesis Generation, utilising an LLM-guided RAG pipeline to screen the policy landscape and output evidence-grounded policy change hypotheses (Aoki, 2024), directly confronting the bottleneck of 'combinatorial complexity'. Here, a policy change hypothesis is defined as an explicit statement of how and why a specific policy is expected to lead to a desired change, outlining a causal pathway and the underlying reasoning for the anticipated outcome. To bridge the divide, the framework proceeds to ③ Scenario Translation, converting these qualitative hypotheses into machine-readable JSON parameters. These serve as inputs for ④ Quantitative Simulation via the SHERPA model, providing the fast, ex-ante feedback necessary to overcome 'slow feedback loops'. Finally, the workflow delivers ⑤ Impact Assessment, transforming the initial broad goals into spatially resolved, quantitatively assessed interventions. This two-phase structure creates a logical and scalable

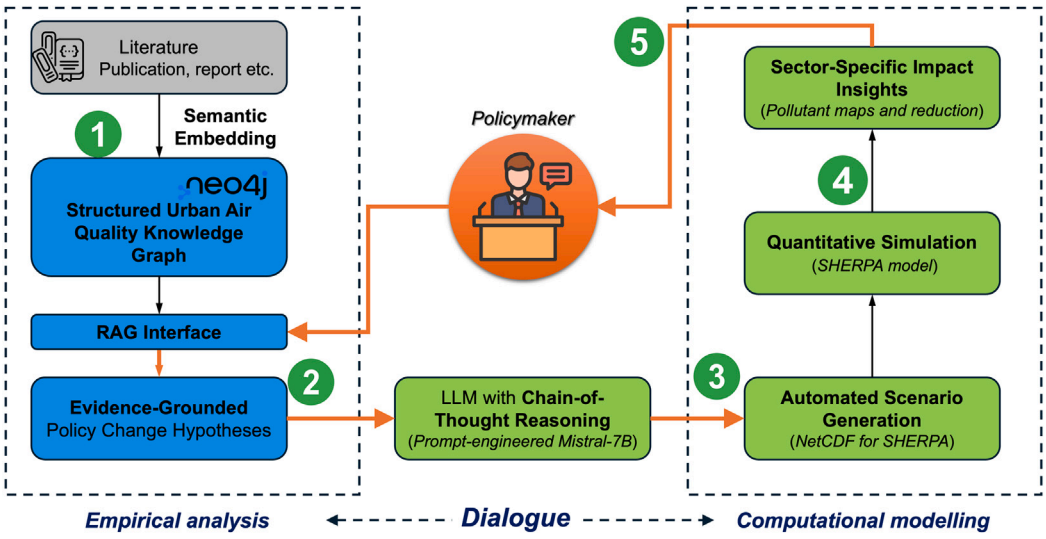


Fig. 2. Dialogue framework for urban air quality evidence synthesis.

workflow that transforms a broad policy goal into a set of quantitatively assessed, actionable interventions, bridging the 'qualitative-quantitative divide' through a formal, automated dialogue between accumulated knowledge and computational simulation.

3.1. Knowledge-grounded hypothesis generation

The first phase of the framework constitutes the 'empirical' component of the dialogue, focused on the systematic extraction, structuring, and operationalisation of domain-specific knowledge from a curated corpus of textual evidence. This is achieved by leveraging Large Language Models (LLMs) to construct a bespoke Urban Air Quality Knowledge Graph (KG) through ontology-guided extraction (Section 3.1.1), and subsequently employing a Retrieval-Augmented Generation (RAG) interface to query this structured evidence (Section 3.1.2).

3.1.1. Knowledge graph development

The credibility and utility of any knowledge-based system are contingent on the quality and scope of its underlying information. Therefore, a curated corpus of authoritative scientific and policy documents is assembled. This corpus was founded on a targeted search of the Web of Science database using keywords specific to UK urban air quality policy, including 'pollutants', 'emission sources', and 'mitigation strategies'. This systematic query yielded 749 peer-reviewed publications. To ensure the knowledge base was not only empirically robust but also directly relevant to the contemporary policy context, the corpus was augmented with structured content from official government reports, including the UK's annual 'Air Pollution in the UK' assessments and the Department for Environment, Food & Rural Affairs' (Defra) summary statistics on national emissions (Płonka et al., 2025). This dual-source approach ensures a comprehensive and balanced foundation for knowledge extraction.

Unlike traditional corpus collection exercises that rely on manual screening, this study employed a so-called 'Extraction-as-Filtering' approach. The entire corpus was subjected to the knowledge extraction pipeline. The relevance of individual documents was determined dynamically by the ontology constraints: texts that did not contain information mapping strictly to the defined classes or relationships resulted in null outputs (i.e., empty JSON objects). This ensured that the KG was populated exclusively with data relevant to the domain.

To transform this vast repository of unstructured text into a structured, queryable asset, a predefined ontology for urban air quality was developed using LinkML. The development of this ontology was driven by a gap analysis of existing semantic standards. While established ontologies such as the Semantic Sensor Network (SSN) focus on real-time observation streams (Dritsas et al., 2025), and the Exposure Ontology (ExO) models biological receptors and health outcomes (Vámos et al., 2024), neither provide the necessary schema to link policy interventions to emission sources. This ontology was developed to map the 'Source-Mitigation' domain specifically. It structures the domain around five primary classes: *Pollutant*, *PollutionSource*, *MitigationMeasure*, *MeteorologicalFactor*, and *StreetCanyonFactor*, as shown in Fig. 3. Each class has specific attributes: a *Pollutant* is defined by its name and category (e.g., Particulate Matter, Gaseous), a *PollutionSource* has a name and a *source_type* (e.g., Mobile Source, Stationary Source), and a *MitigationMeasure* is defined by its name (e.g., Boiler Replacement Program) and a description outlining the intervention. The ontology also captures key contextual factors that influence pollutant dispersion. The *MeteorologicalFactor* class includes properties such as wind speed, temperature, and humidity, which are critical for atmospheric chemistry processes. The *StreetCanyon* class represents features of the built environment, with attributes like height-to-width ratio and orientation relative to wind direction, which significantly affect pollutant concentrations at a local level. The ontology also specifies the relationships between these classes: a *PollutantSourceRelation* connects a Pollutant to a *PollutionSource*, and a *SourceMitigationRelation* links a *PollutionSource* to a *MitigationMeasure*. Similarly, *MeteorologicalDispersionRelation* and *StreetCanyonDispersionRelation* describe how environmental factors influence pollutants. This structured approach enables complex, semantically-aware queries, such as identifying all mitigation strategies relevant to industrial sources of PM2.5 under specific meteorological conditions, thereby transforming unstructured text into a queryable knowledge base.

The role of this ontology is not as a passive container but as an active blueprint that governs the knowledge extraction process. Its predefined schema constrains the task of the LLM, directing it to identify entities that fit the specified classes and to establish connections based on the defined relationships (Caufield et al., 2024). This ontology-guided approach ensures structural consistency and enhances the reliability of the final KG. Accordingly, knowledge extraction was performed using the locally deployed Mistral-7B LLM, which parsed the unstructured texts from the corpus and converted them into structured triples conforming to the LinkML ontology. With the extended context window of 32,000 tokens (approx. 24,000 words), the LLM ensures that the documents can be processed at the chapter or section level. Instead of splitting text into small, arbitrary fragments, this capacity guarantees the LLM maintained access to long-range semantic dependencies. Meanwhile, the ontology alignment was enforced through two

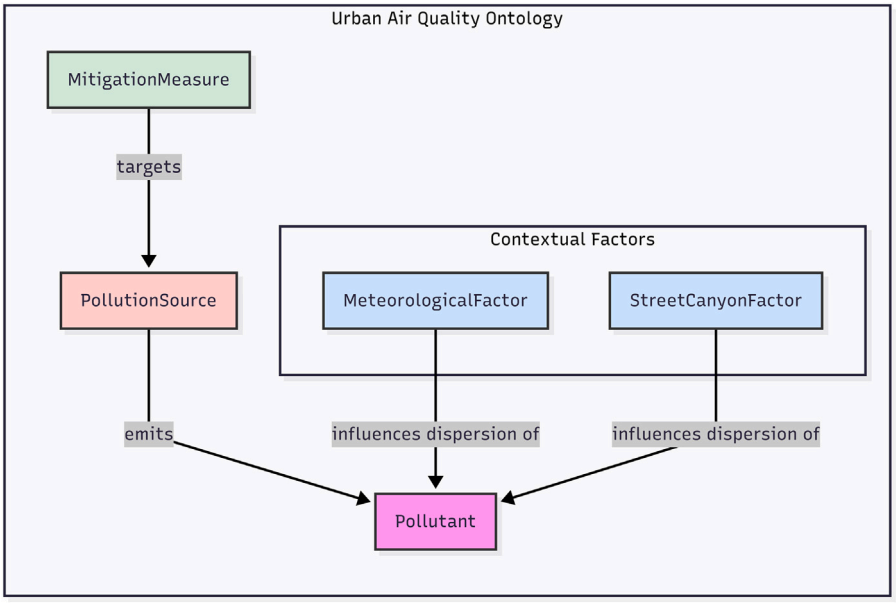


Fig. 3. Conceptual diagram of the urban air quality ontology.

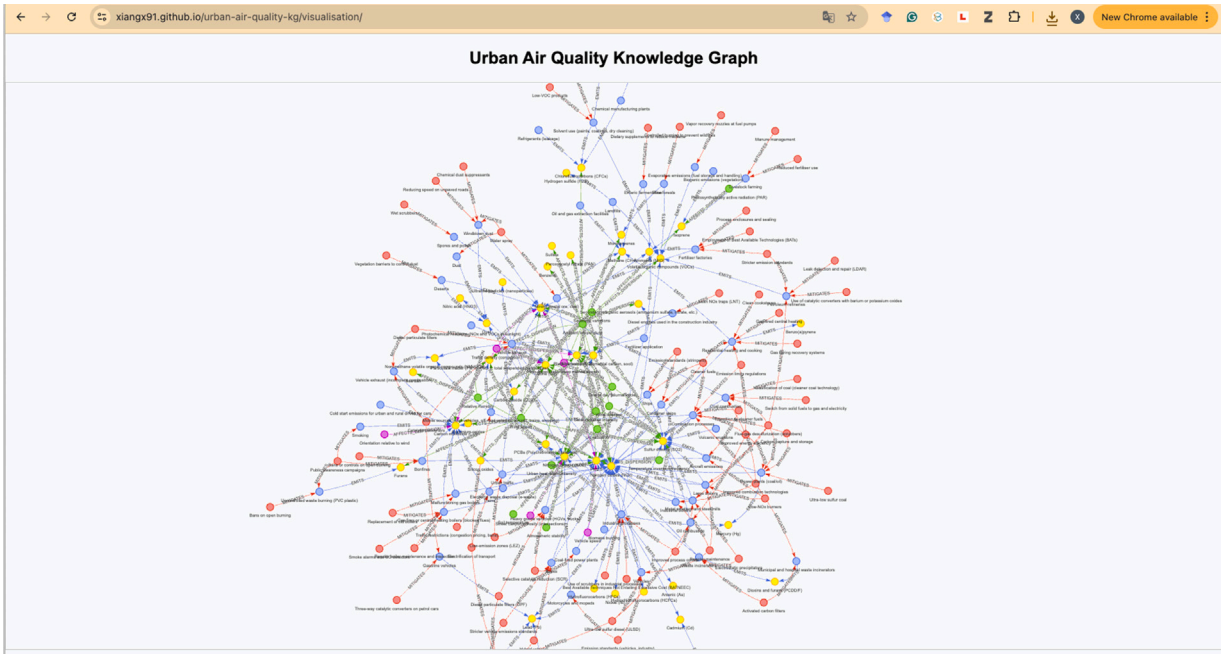


Fig. 4. Interactive visualisation of the urban air quality KG.

steps. The extraction prompt was dynamically constructed to include the full definitions of the ontology (Fig. 3). By embedding the allowed classes and relationships directly into the model's context window, the LLM was conditioned to function as a schema-compliant parser. Meanwhile, a deterministic validation script was applied to the outputs. This post-processing step verified that every extracted triple strictly conformed to the ontology's topology, automatically rejecting any undefined entity types or invalid relationship structures before they could populate the KG. The resulting KG, implemented in a Neo4j graph database, serves as a comprehensive, machine-readable repository of causal and relational knowledge in the domain. In a commitment to open science and transparency, the resulting KG is publicly accessible via an interactive web interface at: https://xiangx91.github.io/urban-air-quality-kg/visualisation/ (see Fig. 4).

A statistically robust assessment of the Knowledge Graph was conducted using a three-stage validation protocol (detailed in Supplementary Material 1). In Stage 1, 150 triples were randomly sampled and independently assessed by two domain experts using a four-category schema: (a) Supported, (b) Supported but indirect/qualified, (c) Unsupported, and (d) Unclear due to wording. Across these paired ratings, the experts achieved a raw agreement of 79.33% and a Cohen's kappa (κ) of 0.6165, indicating 'substantial agreement' beyond chance according to the Landis–Koch scale. Borderline triples identified as indirect or unclear (67 out of 150) were subsequently entered into a Stage 2 adjudication to distinguish valid causal pathways from spurious correlations. This analysis revealed that 100% of these cases were ultimately salvageable: 29 were classified as mediated or context-dependent but valid relationships, and 38 were found recoverable through minor

reformulation of abstract objects or compressed causal chains. This validation profile confirms the Knowledge Graph's reliability as a strategic 'plausibility filter' for foundational domain knowledge, providing the evidence-grounded hypotheses necessary to initialise the subsequent simulation phase. While these results demonstrate that LLM-generated knowledge graphs provide a reliable baseline for managing high-level combinatorial complexity, the need for an adjudication phase confirms that human-in-the-loop oversight remains essential to ensure scientific fidelity, even as the total expert workload is significantly reduced compared to traditional manual construction of KG.

The KG presented in this study constitutes Version 0, serving as the foundational baseline. While this version is static for the current evaluation, its long-term utility for ongoing governance depends on its capacity to evolve as scientific evidence or socio-technical data, such as policy enforcement costs and detailed sociodemographic variables, become available. To facilitate this, a dedicated 'Knowledge Enrich and Validation' workflow[3] is provided in the GitHub repository, allowing domain experts to manually merge new knowledge and revise triples in the local Neo4j database. To further professionalise this lifecycle, we are transitioning to a delta-based versioning model, with RDF-star extensions as the primary focus for future work. This standard supports statement-level annotations, including provenance and temporal metadata, enabling reconstruction and auditing of any past version of the KG with a single query. This formal versioning system is intended to support future evaluation of how the maturation of the knowledge base affects the stability of policy recommendations over time.

3.1.2. RAG-based query process

The use of LLMs in high-stakes public domains introduces significant challenges related to transparency and accountability, as their reasoning processes are often opaque (Pappachan et al., 2024). To ensure that all outputs are verifiable, interpretable, and legally justifiable, this framework adopts a Retrieval-Augmented Generation (RAG) architecture (Ren et al., 2025; Hussien et al., 2025). Unlike standard LLMs that generate responses based on internal parameters, RAG retrieves relevant information from the generated Urban Air Quality KG. This process grounds the LLM's output in verifiable facts, mitigating the risk of factual inaccuracies or 'hallucinations' and transforming the LLM into an auditable decision-support tool. Drawing on recent research across diverse domains (Song et al., 2025; Arazzi et al., 2025), responses from RAG are inherently stable and consistent due to the deterministic nature of the retrieval process when grounded in a structured KG. Because the KG structure and semantic node embeddings are invariant components of the database rather than the LLM backbone, the tests using Mistral-7B and Llama-2-13B confirmed that the retrieved facts remained consistent across various runs. This architectural choice makes using a locally deployed lightweight Mistral-7B model (*mistral-7b-instruct-v0.2.Q4_K_M.gguf*) advantageous. By grounding the model's reasoning in retrieved knowledge, the system achieves high performance and efficiency, ensuring data privacy, low latency, and cost-effectiveness without sacrificing output quality.

To enable effective retrieval, a graph-enhanced retrieval strategy is adopted that integrates dense vector search with structural graph traversal. First, semantic embeddings were computed for each node in the KG using the *all-mpnet-base-v2* model from the Sentence Transformers library. During the retrieval phase, Neo4j's Hierarchical Navigable Small World (HNSW) index is utilised to calculate the Cosine Similarity between the user's query vector and the node vectors (Yang and Kim, 2025). This enables the identification of conceptually related entities even in the absence of exact keyword matches. For instance, a query concerning 'vehicle emissions' can retrieve nodes for 'vehicle exhaust' and 'cold start emissions', demonstrating an ability to infer contextual relevance beyond lexical overlap. To transcend the limitations of simple document retrieval, a structural context expansion step is employed. For every retrieved node, the system executes a Cypher query to traverse its immediate graph neighbourhood, retrieving connected relationships and attributes. This allows the RAG interface to capture precise structured data, such as specific emission factors stored as edge properties or regulatory categories linked to a source.

Furthermore, an evaluation was designed to test the combined capacity of RAG and KG to generate reasonable results for queries typically requiring complex reasoning, rather than assessing a standalone RAG pipeline. The benchmark 20 ground-truth policy proposals were systematically constructed to span established environmental policy instrument types (regulatory, market-based, planning/infrastructure, and behavioural), key air pollutants (PM2.5 and NOx), various governance levels (national, regional, and local), and diverse policy outcomes including concentration reduction, exposure reduction, equity, and co-benefits. This evaluation aligns with contemporary OECD environmental policy standards and peer-reviewed methodologies for measuring policy stringency and sustainable urban governance (Longato et al., 2024; Kruse et al., 2022). The comparative analysis (detailed in Supplementary Material 2) confirms high consistency between the generated hypotheses and expert consensus across the technical and scientific dimensions of air quality policy. The limitations emerge primarily when navigating the intricate socio-technical and governance domains, such as the institutional enforcement capacity, the implementation of real-time monitoring triggers for site shutdowns, and the design of targeted equity-based exemptions for vulnerable populations. These deficits highlight the opportunity to further integrate socio-technical and governance-specific data into the KG to better capture the indirect relationships inherent in policy implementation.

3.2. Quantitative hypothesis testing and simulation

The second phase of the framework constitutes the 'model' component of the dialogue, subjecting the generated policy change hypotheses to rigorous quantitative testing. This is accomplished through an automated pipeline that translates the natural language hypotheses into machine-readable scenarios and executes them within the SHERPA (Screening for High Emission Reduction Potential on Air) simulator. This phase provides the fast, ex-ante feedback necessary to overcome the 'slow feedback loops' that constrain traditional policy evaluation cycles.

Translating high-level policy change hypotheses into the precise, quantitative inputs required by simulation models is typically a manual, time-consuming, and error-prone process. This framework introduces an automated translation pipeline that uses advanced prompt engineering to guide the locally deployed Mistral-7B LLM in converting the policy hypothesis text into a validated JSON object compatible with the SHERPA model's input. At the core of this pipeline is a meticulously engineered master prompt that incorporates several state-of-the-art techniques to ensure accuracy, reliability, and auditability. The architecture of this prompt includes the following components:

- **Role Priming**: The prompt instructs the LLM to adopt the persona of an 'expert policy analyst'. This constrains the model's behaviour, focusing its vocabulary and reasoning patterns within a professional and analytical context.
- **Chain-of-Thought (CoT) Reasoning**: The prompt compels the model to deconstruct the policy text in a step-by-step manner and include this explicit CoT reasoning chain in its output (Wei et al., 2022). Specifically, the prompt guides the model through a specific analytical sequence that involves identifying the core policy action, identifying the target pollutants, mapping the action to the appropriate GNFR sector code, and determining the quantitative reduction from the text. This explicit, structured decomposition makes the model's logic transparent and interpretable to a human analyst, which is a critical feature for any tool intended for urban governance.

[3] https://github.com/XiangX91/urban-air-quality-kg/blob/main/notebooks/Knowledge%20enrich%20and%20validation.ipynb.

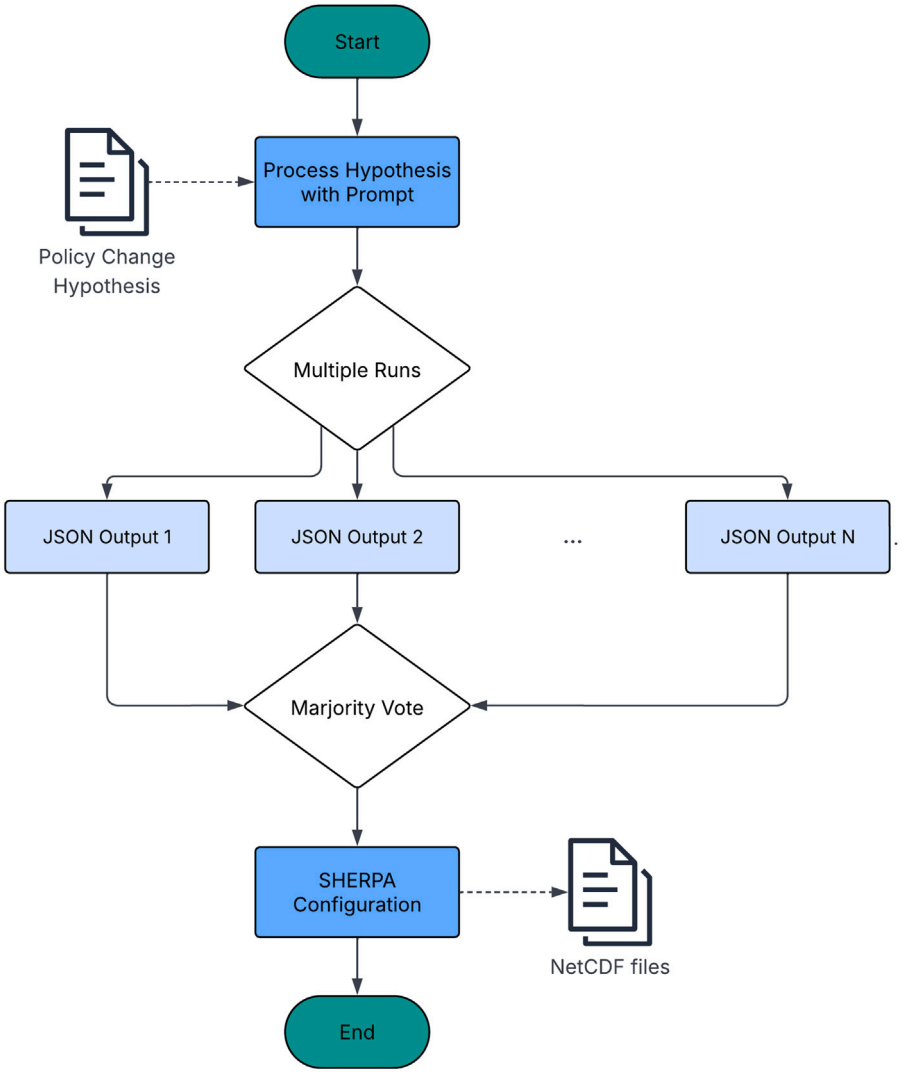


Fig. 5. Workflow of the automated translation pipeline and the self-consistency protocol implemented.

- **Contextual Grounding**: To prevent factual errors and simplify the task, the prompt injects essential domain-specific information directly into the context window. This includes the official Gridded Nomenclature for Reporting (GNFR) sector list used by the SHERPA model. This transforms the task from one of open-ended knowledge recall to a more constrained and reliable mapping exercise, where the model must map a natural language description (e.g., 'passenger cars') to a specific sector code (e.g., 'F_Road Transport').
- **Few-Shot Prompting**: The prompt provides several complete input-to-output examples. This allows the model to learn the precise JSON structure and logic required for the output through imitation, significantly improving the reliability of the generated format.

To ensure robustness against the inherent stochasticity of LLMs, a self-consistency protocol is implemented (Schoenegger et al., 2024), as shown in Fig. 5. For each policy hypothesis, the same master prompt is sent to the LLM multiple times (e.g., $N = 5$) to generate an ensemble of potential JSON outputs. The outputs are then aggregated using a Majority Vote algorithm designed for numerical simulation parameters. The numerical values (i.e., reduction percentages) are grouped, and the most frequent value is selected as the final input for the model. This 'ensemble' approach significantly reduces the error rate of the extraction compared to a single greedy decoding pass, ensuring that the SHERPA simulation is driven by a stable and consensus-based interpretation of the policy text.

The JSON object produced by this pipeline serves as the direct, machine-readable instruction for configuring the SHERPA simulation. This output informs the generation of simulation-ready artefacts, most importantly through the reduction matrix. This matrix explicitly defines the emission reduction percentages for each pollutant precursor and economic sector targeted by the policy, conforming to the specific format required by SHERPA.

The choice of the SHERPA model was guided by the principle of methodological 'fit-for-purpose' design. The framework uses the SHERPA model, a reduced-form tool developed by the European Commission's Joint Research Centre, as a screening tool for rapidly exploring policy scenarios, delivering fast responses to support decision-making (Thunis et al., 2016). Relying on the pre-computed source–receptor relationships (SRRs), SHERPA avoids the need for computationally expensive, full-physics simulations for each new scenario. Crucially, SHERPA is distributed with a comprehensive suite of pre-computed baseline datasets, including spatially resolved emission inventories and meteorological fields derived from full chemical transport model simulations. This eliminates the need for users to generate their own emission inventories or run complex meteorological models. Consequently, the primary input required is the reduction matrix produced

```
🔍 Question explicitly asked: Given the success of the Ultra Low Emission
Zone (ULEZ) in reducing road transport emissions, what are the most
impactful next-generation policies for the London FUA to make further
progress towards meeting WHO guidelines for PM2.5 and NOx?

📖 Answer from local LLM:

Beyond road transport, analysis of the knowledge graph identifies three
priority sectors for reducing NOx and PM2.5 emissions in the London FUA:
Stationary Combustion from Heating: Recent studies show that gas boilers
are now the single largest source of NOx emissions in central London,
accounting for up to 72% of local NOx. Policy interventions should focus
on accelerating the transition from gas boilers to low-emission
alternatives like heat pumps, particularly for commercial and non-domestic
buildings which are the main contributors in central London.
Non-Road Mobile Machinery (NRMM) in Construction: The construction
industry is a significant source of PM2.5 in London, contributing up to 15%
of local emissions. An effective intervention would be to model the impact
of enforcing and accelerating the existing NRMM Low Emission Zone
standards, which mandate a move to Stage IV from 2025 and Stage V by
2030.
Solvent and Product Use: As emissions from other sectors fall, the
relative contribution of Non-Methane Volatile Organic Compounds (NMVOCs)
from solvents has increased significantly. These are key precursors to
ozone and secondary particulate matter. Policies should target a reduction
in NMVOCs through the promotion of low-VOC products and enhanced
enforcement of the Solvent Emissions Directive in commercial settings.

📌 Explicitly retrieved sources from Neo4j:
-
name: Industrial boilers
category: StationarySource
-
name: Residential heating and cooking
category: AreaSource
-
name: Malfunctioning gas boilers
category: StationarySource
-
name: Non-Road Mobile Machinery
category: MobileSource
-
name: Emissions from Construction
category: AreaSource
-
name: Solvent use (paints, coatings, dry cleaning)
category: AreaSource
-
name: Evaporative emissions (fuel storage and handling)
category: AreaSource
```

Fig. 6. Example of a natural language query answered by the RAG interface.

by the automated translation pipeline, a set of perturbation coefficients applied to these pre-existing baselines, along with an area mask to delineate the study's geographic scope.

Applying the reduction matrix to these baseline datasets, the SHERPA model computes the resulting changes in pollutant concentrations. The primary outputs of the simulation are spatially resolved NetCDF result files. These files provide detailed data on the policy's impact and are designed for further analysis and visualisation. They allow for the generation of impact maps in Geographic Information Systems (GIS) and the calculation of key metrics such as the 'relative potential', defined as the proportionate change relative to baseline concentrations. This quantitative output provides the auditable, ex-ante feedback on potential policy impacts that is essential for adaptive and evidence-informed governance.

3.3. System requirements and reproducibility

Computationally speaking, the framework is designed to be accessible to policymakers and relevant stakeholders with limited resources. The computational requirements were benchmarked on professional-grade hardware. The KG development and RAG inference were executed using a locally deployed Mistral-7B model. For this study, a single NVIDIA RTX 6000 Ada Generation (48 GB) was utilised, achieving an inference speed of 135 tokens/second. But the framework can operate on more modest hardware. The minimum recommended configuration requires a GPU with 24 GB of VRAM (e.g., NVIDIA RTX 3090) to support the model weights and the full 32k token context window. The Neo4j graph database and the SHERPA simulation model are computationally lightweight and run efficiently on standard system RAM (16 GB+) and CPUs (e.g., Intel Core i7/i9), with typical simulation scenarios completing in under 60 s.

4. Case study

This case study demonstrates the application of the proposed evidence synthesis framework to a realistic and complex policy challenge in the London Functional Urban Area (FUA). The objective is to validate the framework's ability to translate high-level strategic goals into specific, evidence-based policy interventions and to assess their impact on urban air quality quantitatively.

London has made significant strides in improving air quality, mainly by targeting road transport emissions through policies like the Ultra Low Emission Zone (ULEZ). The London-wide expansion of the ULEZ in 2023 has resulted in over 95% of vehicles meeting the required emission standards, contributing to a dramatic reduction in transport-related pollutants. Recent evidence from the 2025 London-wide ULEZ report indicates that by 2024, NO_2 concentrations across the capital were approximately 27% lower than they would have been without the scheme (Greater London Authority & Transport for London, 2025). This builds on earlier gains, where NOx emissions had already fallen by 31% between 2016 and 2019.[4]

This success has shifted the policy landscape. As transport's contribution diminishes, other previously less prominent emission sources now represent a larger proportion of the remaining problem. Currently, London still exceeds the World Health Organisation (WHO) guidelines for both NO_2 and PM2.5, necessitating a new generation of more targeted policies. This section demonstrates the proposed tool's ability to help a policymaker navigate this complex environment and formulate an effective, multi-pronged strategy for the London FUA.

4.1. Hypothesis generation: Identifying post-ULEZ policy priorities for London

The generation of the policy change hypothesis is initiated when a policymaker poses a high-level strategic query to the RAG-enabled knowledge graph interface. To demonstrate the framework's ability to navigate a complex, real-world policy environment, this case study begins with the following query:

> *'Given the success of the Ultra Low Emission Zone (ULEZ) in reducing road transport emissions, what are the most impactful next-generation policies for the London FUA to make further progress towards meeting WHO guidelines for PM2.5 and NOx?'*

Upon receiving this query, the RAG deconstructs the user's intent, recognising the key constraints: the proposed policy hypotheses must be 'next-generation', addressing sources beyond the primary focus of the ULEZ, and they must target both PM2.5 and NOx. The RAG then queries its knowledge graph, using semantic search to find nodes related to significant non-transport emission sources in London and their corresponding mitigation strategies. Rather than returning a generic list of interventions, the RAG synthesises the retrieved information into coherent, evidence-based policy hypotheses, identifying three priority areas for intervention, as shown in Fig. 6.

The RAG first identified stationary combustion from heating as a primary area for intervention. This aligns with recent source apportionment studies, which reveal that as road transport emissions decline, gas combustion in boilers has become the single largest source of NOx in central London, accounting for up to 72% of local emissions (Cliff et al., 2025). The RAG interface corroborated the significance of this sector by successfully retrieving highly relevant nodes from the knowledge graph, including 'Industrial boilers' and 'Residential heating and cooking'. This successful grounding demonstrates the framework's capacity to connect a high-level strategic goal to specific, verifiable emission sources, thereby forming the first pillar of the proposed intervention strategy.

The second policy hypothesis targets emissions from Non-Road Mobile Machinery (NRMM), a sector identified as a major contributor to particulate pollution. London's extensive construction activity, in particular, is responsible for up to 15% of local PM2.5 emissions (Lončarević et al., 2022). The framework consequently proposes accelerating the enforcement of the London NRMM Low Emission Zone, which mandates a tightening of standards to Stage IV from 2025 and Stage V from 2030. This recommendation was substantiated by the retrieval of nodes such as 'Non-Road Mobile Machinery' and 'Emissions from Construction', linking the policy directly to its source category and documented reduction potential.

Finally, the RAG proposed a third hypothesis targeting solvent and product use, demonstrating its ability to identify less-publicised but increasingly critical emission sources. As other primary sources are abated, the relative contribution of Non-Methane Volatile Organic Compounds (NMVOCs) from solvents has grown, now becoming a dominant precursor source for secondary organic aerosol and other pollutants (Lewis et al., 2020; Mo et al., 2021). The resulting policy recommendation, aimed at enhancing the enforcement of the Solvent Emissions Directive (SED), was directly informed by retrieving the 'Solvent use (paints, coatings, dry cleaning)' node from the knowledge graph.

The RAG-guided analysis yielded a multi-sectoral portfolio of qualitative policy hypotheses targeting heating, construction, and solvent use. This first phase successfully translated a broad strategic goal into a concise set of feasible, evidence-based policy concepts, establishing a robust and auditable foundation for the subsequent quantitative hypothesis testing.

4.2. Hypothesis testing: Quantifying the impact of the proposed policies

An automated translation pipeline converts the qualitative policy hypotheses developed in Section 4.1 into a quantitative, machine-readable scenario for the SHERPA simulation. Note that the particular reduction percentage represents the maximum technically feasible reduction potential identified for the GNFR sector. The three policy change hypotheses identified by the RAG are processed by the automated translation pipeline, using the policy text as the direct input to generate the structured JSON objects required by SHERPA (Figs. 7, 8, and 9). Each JSON object demonstrates the successful deconstruction of the qualitative policy text into its core components: a human-readable *scenario_name*, the original *source_policy_text* for traceability, a *reasoning_chain* that makes the model's logic explicit, and a *reductions* array that specifies the quantitative parameters for the simulation.

In Fig. 7, the translation pipeline correctly identifies NOx as the target pollutant for the boiler replacement policy and maps it to the 'GNFR C: Other Stationary Combustion sector', extracting the 72% reduction potential directly from the source text. Similarly, Fig. 8 shows the translation of the NRMM policy into a 15% reduction in PM2.5 for the 'GNFR I: Off-road sector', and Fig. 9 presents the translation of the solvent policy into a 50% reduction in NMVOCs for the 'GNFR E: Solvents sector'.

Once generated and validated, these JSON objects are used to configure a script that produces a single NetCDF (.nc) input file for the SHERPA model. The simulation is then run for the London FUA to quantify the impact of the combined policy package. The primary output of the SHERPA simulation is spatially resolved impact maps, which provide clear, quantitative evidence of the policy package's effectiveness. The impact of the first policy hypothesis, targeting NOx emissions from commercial and domestic boilers, is illustrated in Fig. 10. Assuming full implementation of the policy, the model reveals a considerable reduction in annual mean NOx concentration, an effect that is geographically concentrated in central London. In these high-density commercial and residential areas, the decrease reaches up to 11.5% of NOx, equivalent to an absolute reduction of 3.96 $\mu g\,m^{-3}$. This spatial pattern directly confirms the value of targeting these key stationary emission sources where they are most dense.

The second policy, enforcing stricter emission standards for construction machinery, delivers a targeted reduction in primary PM2.5. The simulation's spatial analysis (Fig. 11) reveals that the impact is

[4] https://www.london.gov.uk/programmes-strategies/environment-and-climate-change/environment-and-climate-change-publications/london-wide-ultra-low-emission-zone-one-year-report.

Source Policy Text:

> *"Stationary Combustion from Heating: Recent studies show that gas boilers are now the single largest source of NOx emissions in central London, accounting for up to 72% of local NOx. Policy interventions should focus on accelerating the transition from gas boilers to low-emission alternatives like heat pumps, particularly for commercial and non-domestic buildings, which are the main contributors in central London."*

Reasoning Chain (Chain-of-Thought):

- **Identify Policy Action**: The policy intervention is to accelerate the transition from gas boilers to low–emission alternatives like heat pumps in commercial and non–domestic buildings.
- **Identify Target Pollutant(s)**: The specific target pollutant mentioned is NOx (Nitrogen Oxides).
- **Map to GNFR Sectors**: Heating in commercial and non–domestic buildings is classified under C_OtherStationaryComb.
- **Determine Quantitative Reduction**: The source text states that the targeted gas boilers account for up to 72% of local NOx, providing a direct, evidence–based figure for the potential reduction.
- **Final Summary**: This scenario models a 72% reduction in NOx from the C_OtherStationaryComb sector, representing the impact of replacing commercial gas boilers.

Simulation Parameters ('reductions'):

```
"reductions": [
  {
    "gnfr_sector_code": "C_OtherStationaryComb",
    "gnfr_sector_description": "Emissions from small combustion
    sectors (e.g., commercial, institutional, residential,
    agricultural)",
    "pollutant": "NOx",
    "reduction_percent": 72
  }
]
```

Fig. 7. Generated JSON object for the heating and combustion intervention.

more localised, with the benefits concentrated in zones of high construction and development activity. The peak reduction in annual mean PM2.5 concentration is 0.256%, equivalent to 0.0243 $\mu g\,m^{-3}$. The value of this outcome lies not in its city-wide magnitude, but in confirming the policy's effectiveness at mitigating emissions directly at their source and lowering exposure in specific, high-activity areas.

The third policy targets NMVOCs from solvent use, a key precursor to the secondary organic aerosols (SOA) that are a major component of PM2.5. The simulated impact of this intervention (Fig. 12) reveals the complex and non-linear nature of atmospheric chemistry. Rather than the expected benefit, the model predicts that this well-intended policy would result in slight increases in PM2.5 across the London FUA, with a peak increase of 0.446%, equivalent to an absolute rise of 0.0329 $\mu g\,m^{-3}$.

This seemingly counterintuitive result is a well-documented atmospheric chemistry phenomenon. In high-NOx urban environments like London, NOx can suppress the formation efficiency of SOA from certain VOCs. Reducing NMVOCs alters the complex chemical balance, paradoxically increasing the formation efficiency of other secondary aerosols, and leading to the minor, localised increases in PM2.5 (Li et al., 2024; Wang et al., 2022). The ability of simulation to capture this non-linear response is a critical validation of the two-phase framework. While the grounded policy change hypotheses provide essential strategic direction, the simulation is indispensable for revealing complex effects for the specific case. This finding underscores the need for dialogue between qualitative evidence and simulation in the design of robust and effective air quality management strategies. By connecting knowledge-driven hypothesis generation with quantitative hypothesis testing, the framework empowers policymakers to design, test, and refine more sophisticated and effective air quality strategies for complex urban environments.

5. Discussion

5.1. Limitations of SHERPA and feasibility of machine learning integration

The utilisation of the SHERPA model enables the computational efficiency required to navigate combinatorial complexity, but its results

Source Policy Text:

> *"Non–Road Mobile Machinery (NRMM) in Construction: The construction industry is a significant source of PM2.5 in London, contributing up to 15% of local emissions. An effective intervention would be to model the impact of enforcing and accelerating the existing NRMM Low Emission Zone standards, which mandate a move to Stage IV from 2025 and Stage V by 2030."*

Reasoning Chain (Chain-of-Thought):

- **Identify Policy Action**: The policy is to enforce and accelerate the existing Non–Road Mobile Machinery (NRMM) Low Emission Zone standards for the construction industry.
- **Identify Target Pollutant(s)**: The specific target pollutant identified is PM2.5.
- **Map to GNFR Sectors**: Non–Road Mobile Machinery used in construction and industry is categorized under the I_Offroad sector.
- **Determine Quantitative Reduction**: The text states the construction industry contributes up to 15% of local PM2.5 emissions, providing a clear, evidence–based figure for the reduction potential.
- **Final Summary**: This scenario models a 15% reduction in PM2.5 from the I_Offroad sector, reflecting the impact of stricter standards for construction machinery.

Simulation Parameters ('reductions'):

```
"reductions": [
  {
    "gnfr_sector_code": "I_Offroad",
    "gnfr_sector_description": "Emissions from machinery
    (industry, households, agriculture), railways, and fishing",
    "pollutant": "PM2.5",
    "reduction_percent": 15
  }
]
```

Fig. 8. Generated JSON object for the NRMM intervention in the construction sector.

are subject to the inherent assumptions of reduced-form modelling. Although atmospheric chemistry is physically non-linear, it has been verified that these dynamics can be accurately approximated by a linear function in SHERPA ($\Delta C_j = \sum_p^P \sum_i^N a_{ijp} \Delta E_{ip}$) when averaged over long periods (e.g., annual averages in this paper) if emission reductions do not exceed 50% (Thunis et al., 2018). Therefore, the framework's ability to capture the counterintuitive result in Fig. 12 is due to the complex secondary reaction dynamics that are intrinsically encoded in the Source–receptor Relationship (SRR) coefficients. These SRRs are derived from training datasets generated by high-fidelity chemical transport models that solve non-linear equations for gas-to-particle conversion.

Furthermore, the SHERPA model provides annual-average results suitable for broad ex-ante policy screening, though this temporal aggregation necessarily masks short-term (e.g., daily or hourly) exceedances and building-scale hotspot dynamics. According to the US EPA's tiered assessment approach, a tiered decision framework for model selection can be adopted based on policy characteristics. The current framework with SHERPA acts as a proactive filter for 'Strategic Screening', optimised for urban and regional spatial scales (1–100 km) and long-term targets where its linearised transfer functions are scientifically robust. However, the transition to 'Refined Modelling', which utilises high-fidelity models such as WRF-Chem, is necessary when policies target microscale hotspots (<1 km), require hourly temporal resolution (e.g., rush-hour peak management), or propose deep structural reductions (>50%).

It is also essential to acknowledge that the emission reduction figures extracted by the current framework represent theoretical technical maximums under idealised conditions. To bridge the gap between these ideals and real-world outcomes, the next phase of this framework will integrate a dedicated cost–benefit evaluation layer. This roadmap involves adopting the Social Return on Investment (SROI) framework to systematically measure and monetise the social and environmental value created relative to the capital investment required for enforcement. This mechanism will allow the system to scale theoretical maximums down to realistic compliance scenarios by accounting for multi-dimensional implementation barriers, including institutional enforcement capacity, local market readiness for abatement technologies, and up-front capital costs. This future integration will provide the

Source Policy Text:

> *"Solvent and Product Use: As emissions from other sectors fall, the relative contribution of Non–Methane Volatile Organic Compounds (NMVOCs) from solvents has increased significantly. These are key precursors to ozone and secondary particulate matter. Policies should target a reduction in NMVOCs through the promotion of low–VOC products and enhanced enforcement of the Solvent Emissions Directive in commercial settings."*

Reasoning Chain (Chain-of-Thought):

- **Identify Policy Action**: The policy involves promoting low–VOC (Volatile Organic Compound) products and enhancing the enforcement of the Solvent Emissions Directive for commercial entities.
- **Identify Target Pollutant(s)**: The specific target pollutant is explicitly stated as Non–Methane Volatile Organic Compounds (NMVOCs).
- **Map to GNFR Sectors**: Emissions from the use of solvents and related products are categorized under the E_Solvents sector.
- **Determine Quantitative Reduction**: The text does not provide a specific number but implies a 'significant' reduction. A 50% reduction is assumed, reflecting a strong, combined policy push on both product substitution and regulatory enforcement.
- **Final Summary**: This scenario models a 50% reduction in NMVOC emissions from the E_Solvents sector due to policies promoting low–VOC products and stricter regulations.

Simulation Parameters ('reductions'):

```
"reductions": [
  {
    "gnfr_sector_code": "E_Solvents",
    "gnfr_sector_description": "Emissions from the use of
    solvents",
    "pollutant": "NMVOC",
    "reduction_percent": 50
  }
]
```

Fig. 9. Generated JSON object for the solvent and product use intervention.

'socio-technical anchor' necessary to ensure that policy hypotheses are not just scientifically plausible, but economically implementable.

Beyond conventional chemical transport models and reduced-form models like SHERPA, machine learning techniques, such as Gaussian Process Regression (GPR), offer great potential as high-fidelity surrogates (Wong et al., 2025). While surrogate models demonstrated strong predictive performance, their implementation within a proactive governance framework faces several fundamental practical issues. First, surrogate models face a generalisability-performance trade-off: because they learn statistical associations rather than explicitly solving physical equations, their accuracy highly depends on the local emission, meteorological, and topographical conditions of their training set. It means they may perform poorly in heterogeneous or 'unseen' environments where local sources differ from those in the training ensemble (Zheng et al., 2019). Second, surrogate models cannot reliably predict pollutant levels or chemical regimes that fall outside the range of their training data (Nowack et al., 2021). In contrast, utilising physics-informed SRRs, SHERPA captures complex dynamics, even under novel policy configurations that may not exist in the historical training manifold. Finally, from an engineering perspective, a standardised model like SHERPA supports 'plug-and-play' across 150 European cities because their input taxonomies are uniform. Replacing this with machine learning emulators would require fresh role priming, context grounding, and exhaustive few-shot testing to ensure the semantic parser reliably maps to each model's unique input requirements, imposing significant labour and testing overhead that would undermine the practicality of this work.

5.2. Implications for evidence-informed policymaking

This study proposed and demonstrated a two-phase framework that operationalises the 'dialogue' between qualitative evidence and quantitative simulation for supporting proactive urban policymaking. The fundamental differences between this new paradigm and conventional approaches are summarised in Table 3. Leveraging LLMs to synthesise knowledge from textual sources for policy screening and

Table 3
Comparative analysis of environmental policy analysis paradigms.

Feature	Conventional quantitative models (e.g., SHERPA)	Traditional qualitative analysis	Proposed synthesis framework
Primary data sources	Structured numerical data (e.g., emissions inventories)	Unstructured text (e.g., interviews, case studies)	Hybrid: Structured and unstructured data, real-time sources
Representation of constraints	Implicit or highly simplified (via parameters)	Rich, narrative-based	Explicit, structured, and queryable via knowledge graph
Handling of Uncertainty	Parametric sensitivity analysis	Acknowledged but not formalised	Explored through scenario generation and transparent reasoning
Analytical transparency	Often a 'black box'	Interpretable but not scalable	High, via explainable chain-of-thought reasoning paths
Scalability	High (computationally)	Low (labour-intensive)	High, automated by LLMs
Accessibility for non-experts	Low (requires technical expertise)	Moderate	High (via natural language interface and visualisations)

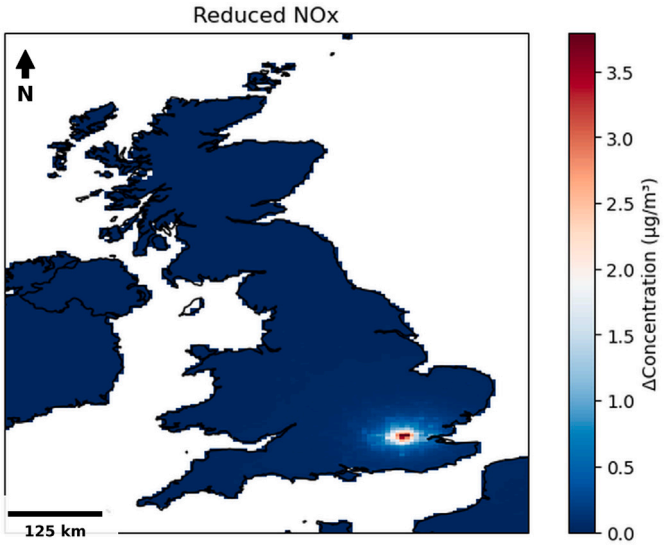


Fig. 10. Simulated reduction in annual mean NOx for the heating and combustion intervention for London FUA.

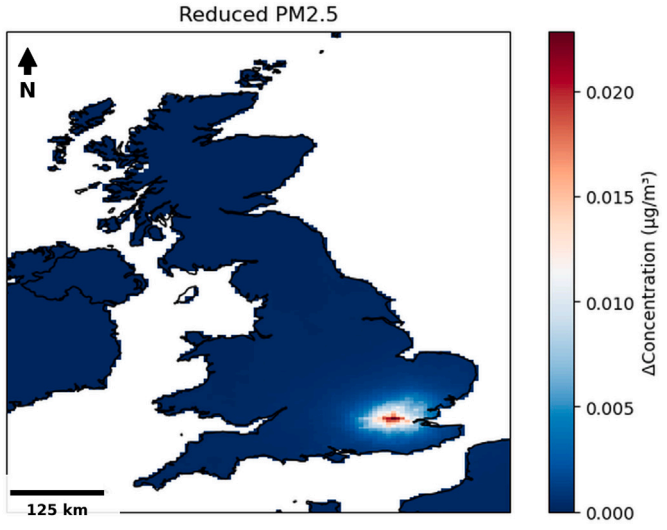


Fig. 11. Simulated reduction in annual mean PM2.5 for the NRMM intervention in the construction sector for London FUA.

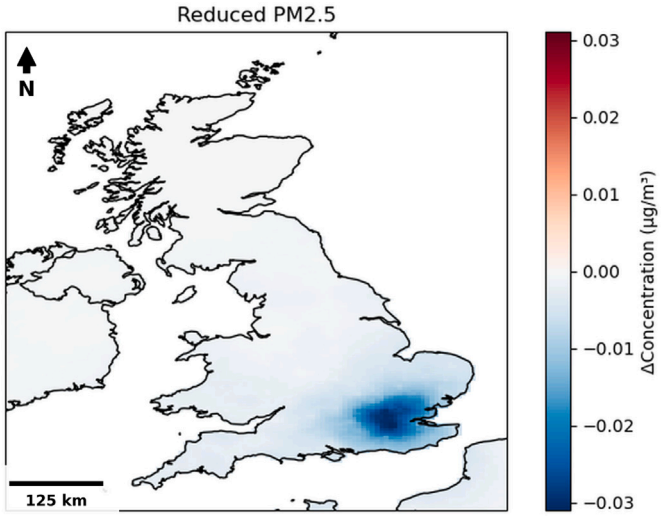


Fig. 12. Simulated impact on annual mean PM2.5 for the solvent and product use intervention for London FUA.

using a model-informed approach for rigorous hypothesis testing, the framework directly addresses the critical bottlenecks of 'combinatorial complexity' and 'slow feedback loops' that hinder conventional policymaking. Traditional qualitative analysis and ex-post evaluation often operate on timelines of years, as determining causal effects requires isolating the specific signal of an intervention from the 'deafening noise' of confounding variables. This timeline is further elongated by the administrative architecture of regulatory science. For example, the Inner London ULEZ One Year Report was published in February 2023, 16 months after the expansion in October 2021. In contrast, the proposed framework compresses this cycle from months to minutes. By replacing longitudinal observation with ex-ante simulation, the system can generate and test a portfolio of scenarios in under 60 min. This acceleration shifts 'waiting for proof' to 'proactive screening', enabling the identification of risks before political capital is committed.

Beyond its technical contributions, the framework presented in this study carries significant implications for promoting procedural equity in urban environmental engineering. Conventional approaches to evidence-informed policymaking are often resource-intensive, requiring specialised expertise that inadvertently marginalises resource-constrained municipalities. By automating the most labour-intensive aspects of evidence synthesis and providing the Urban Air Quality KG (Fig. 5) as a shareable public good, this tool dramatically lowers the operational costs of sophisticated policy screening. This democratisation of advanced analytical capabilities empowers a wider range of cities to move beyond reactive measures and proactively engineer healthier urban environments.

Looking ahead, a critical roadmap for the framework's development is the formal integration of environmental equity, i.e., addressing the

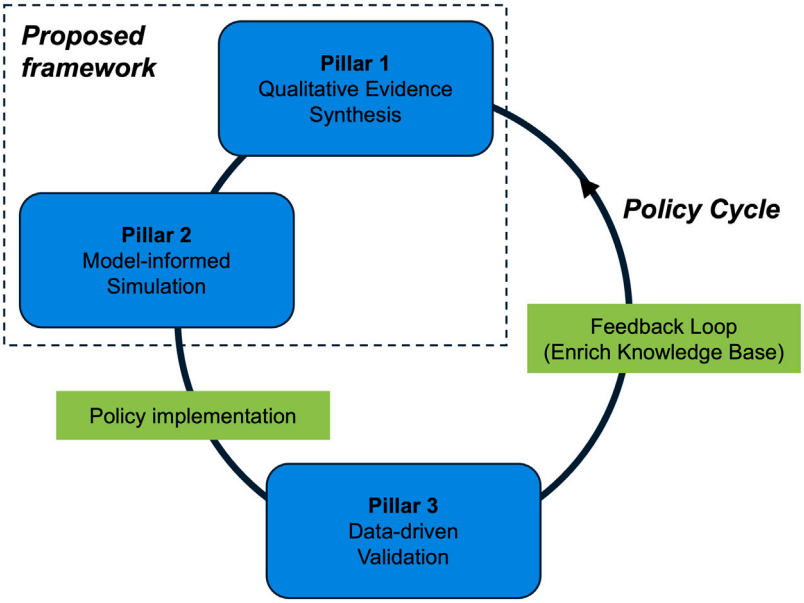


Fig. 13. Three-pillar framework for a continuous evidence cycle.

specific distribution of environmental burdens across socioeconomic and vulnerable groups. The integration of environmental justice requires modifications in both synthesis phases. In the first phase, the KG ontology can be enriched with attributes related to distributive justice, such as Index of Multiple Deprivation (IMD) scores and ethnicity patterns from census data. For the simulation phase, the framework must transition toward high-resolution spatial modelling (<1 km). Research shows that air quality modelling with coarse horizontal grids tends to underpredict exposure disparities, whereas high-resolution analysis can increase population-weighted exposure estimates by up to 27% (Paolella et al., 2018). By outputting spatially explicit metrics for vulnerable groups, such as 'red route' communities living along main traffic arteries, the framework will eventually ensure that urban policy design proactively mitigates, rather than exacerbates, existing environmental inequalities.

5.3. Future vision: Towards a three-pillar framework for a continuous evidence cycle

The two-phase 'dialogue' between qualitative evidence and quantitative simulation demonstrated in this study represents an automation and scaling advance in evidence synthesis. However, it is the foundational component of a more comprehensive vision for a continuous, learning-oriented evidence cycle, as conceptualised in Fig. 13. This conceptual roadmap aims to integrate the full spectrum of the empirical paradigm, creating a tripartite dialogue between three complementary pillars of knowledge: qualitative evidence from text, model-informed simulation, and quantitative data from real-world sensors.

Each pillar offers a unique way of knowing, with strengths that compensate for the limitations of the others:

- **Pillar 1: Qualitative Evidence Synthesis (The 'What Could We Do?').** This pillar, which corresponds to the first phase of the proposed current framework, uses LLM-guided analysis of textual sources to provide generalisable, high-level causal knowledge. It captures the accumulated wisdom and nuanced understanding of domain experts, making it the ideal instrument for overcoming combinatorial complexity and generating plausible, evidence-grounded hypotheses for policy screening.
- **Pillar 2: Model-Informed Simulation (The 'What Would Happen If?').** This pillar, corresponding to the second phase, provides the engine for robust, ex-ante causal testing of the generated hypotheses. As demonstrated by the case study's discovery of non-linear atmospheric chemistry effects, simulation is uniquely capable of uncovering complex system dynamics and potential unintended consequences before a policy is implemented.
- **Pillar 3: Data-Driven Validation (The 'What Actually Happened?').** This third, future pillar introduces real-world observation from sensor networks to complete the evidence cycle. While the knowledge embedded in text is generalisable and curated, sensor data is objective, high-resolution, and context-specific. Its primary limitation is its ex-post nature: it can only measure the world as it is, not as it might be. Its power, therefore, lies in its complementary role as the ultimate arbiter of both model predictions and the knowledge base itself.

Envisioning a fully integrated cycle reveals the optimal synergy between these approaches. The policymaking process would begin with Pillar 1 to generate high-potential hypotheses, which are then tested using Pillar 2 to provide rapid, ex-ante feedback. Following the implementation of a selected policy, Pillar 3 would be deployed to collect real-world data from sensor networks. This data is used to validate or challenge the model's predictions and can feed back into the Urban Air Quality KG.

While this tripartite integration remains a forward-looking vision and has not yet been empirically tested, the intended positive feedback loop from Pillar 3 back to Pillar 1 serves as the primary roadmap for future research. Two complementary mechanisms are promising to close this loop. First, LLM-enabled Time-Series-to-Text (TS2T) generation can be utilised to bridge the semantic gap between raw sensor signals and structured knowledge. In this vision, an LLM agent monitors data streams to translate statistical anomalies into narrative 'synthetic field reports' (e.g., 'Sustained PM2.5 elevation detected during non-commuting hours, potentially linked to domestic heating') (Xie et al., 2025). These narratives can then be parsed to instantiate new nodes within the KG, dynamically expanding its scope to include episodic real-world evidence (Hou et al., 2025). Second, the application of automated causal discovery algorithms to observational time-series is intended to mine latent dependency structures (Yang et al., 2024). This approach enables the autonomous proposal of new relationship edges or the refinement of confidence weights whenever empirical data-driven links are detected. By positioning this cycle as future work,

we acknowledge that substantial technical development in sensor-to-knowledge integration is required before such a system can be operationalised, thereby providing a clear trajectory for the subsequent evolution of proactive urban environmental engineering.

6. Conclusion

While a fundamental divide exists between the vast, unstructured knowledge in textual evidence and the predictive capabilities of quantitative simulation, it is primarily maintained by the intensive manual labour and specialised expertise traditionally required to bridge them. This study introduces and validates a two-phase framework that accelerates the 'dialogue' between these evidence paradigms to address this challenge. By coupling a Retrieval-Augmented Generation (RAG) pipeline that queries a bespoke knowledge graph with an automated simulation pipeline, the framework addresses the critical bottlenecks of 'combinatorial complexity' and 'slow feedback loops' that hinder conventional policymaking. The first phase distils a vast body of unstructured knowledge into a manageable set of evidence-grounded policy hypotheses. The second phase subsequently translates these hypotheses into machine-readable scenarios for rapid, ex-ante quantitative testing.

The London case study demonstrates the framework's practical utility in navigating a complex, post-ULEZ policy landscape. The RAG pipeline identified a portfolio of next-generation interventions targeting heating, construction, and solvent use. Subsequent simulations with the SHERPA model produced spatially resolved impact assessments for each policy. Crucially, the simulation phase revealed complex system dynamics that would be invisible to qualitative analysis alone. The discovery of a counterintuitive, localised increase in harmful particulate matter (PM2.5) due to the solvent reduction policy, which is a consequence of non-linear atmospheric chemistry in a high-NOx environment, provides robust validation of the framework.

For policymakers, this research offers a scalable and transferable tool for more proactive and intelligent urban governance. The open-source nature of the knowledge graph and simulation code provides a foundation for other cities to adapt this methodology to their own unique contexts and challenges. By transforming fragmented information into clear, decision-relevant insights, the framework empowers cities to systematically screen a wide range of potential interventions and align policy design with public health and environmental goals.

Looking ahead, the two-pillar framework forms the foundation for a more comprehensive, three-pillar vision for a continuous evidence cycle. The integration of real-time sensor data as a third pillar will close the loop between hypothesis, simulation, and real-world observation. This will create a dynamic, learning-oriented system where sensor data is used to validate model forecasts and continually refine the knowledge base. This integrated approach, blending expert insights from text, predictive power from simulation, and validation from observation, offers a robust pathway for navigating the complexity of urban problems far beyond air quality, including climate resilience, mobility planning, and public health.

CRediT authorship contribution statement

Xiang Xie: Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mohamad Kassem:** Writing – review & editing, Visualization, Supervision, Methodology. **Sheen Mclean Cabaneros:** Writing – review & editing, Visualization, Validation, Methodology. **Haixiao Pan:** Writing – review & editing, Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jenvman.2026.129470.

Data availability

The source code for constructing and retrieving the Urban Air Quality Knowledge Graph and simulating is open-source and publicly available on GitHub at https://github.com/XiangX91/urban-air-quality-kg and https://github.com/XiangX91/urban-air-mitigation-sim , respectively. The baseline datasets for the SHERPA model, including emission inventories and source–receptor models, are distributed with the tool and are derived from publicly maintained European environmental data repositories.

References

Amann, M., Kiesewetter, G., Schöpp, W., Klimont, Z., Winiwarter, W., Cofala, J., Rafaj, P., Höglund-Isaksson, L., Gomez-Sabriana, A., Heyes, C., et al., 2020. Reducing global air pollution: the scope for further policy interventions. Phil. Trans. R. Soc. A 378 (2183), 20190331.

Aoki, G., 2024. Large language models in politics and democracy: A comprehensive survey. arXiv preprint arXiv:2412.04498.

Arazzi, M., Marconi Sciarroni, M., Nocera, A., Storti, E., 2025. RAG-IoE: IoT context-aware information retrieval with large language models in industry 5.0. ACM Trans. Internet Things 6 (4), 1–31.

Assaf, G., Hu, X., Assaad, R.H., 2023. Mining and modeling the direct and indirect causalities among factors affecting the urban heat island severity using structural machine learned Bayesian networks. Urban Clim. 49, 101570.

Banitz, T., Schlüter, M., Lindkvist, E., Radosavljevic, S., Johansson, L.-G., Ylikoski, P., Martínez-Peña, R., Grimm, V., 2022. Model-derived causal explanations are inherently constrained by hidden assumptions and context: The example of baltic cod dynamics. Environ. Model. Softw. 156, 105489.

Carodenuto, S., Ziga-Abortta, F.R., Sotirov, M., 2024. External Europeanization through timber trade agreements: Tracing causality in environmental governance reform. Political Geogr. 109, 103065.

Caufield, J.H., Hegde, H., Emonet, V., Harris, N.L., Joachimiak, M.P., Matentzoglu, N., Kim, H., Moxon, S., Reese, J.T., Haendel, M.A., et al., 2024. Structured prompt interrogation and recursive extraction of semantics (SPIRES): A method for populating knowledge bases using zero-shot learning. Bioinformatics 40 (3), btae104.

Cliff, S.J., Drysdale, W., Lewis, A.C., Møller, S.J., Helfter, C., Metzger, S., Liddard, R., Nemitz, E., Barlow, J.F., Lee, J.D., 2025. Evidence of heating-dominated urban NOx emissions. Environ. Sci. Technol. 59 (9), 4399–4408.

Coeckelbergh, M., 2025. LLMs, truth, and democracy: An overview of risks. Sci. Eng. Ethics 31 (1), 4.

Cooke, S.J., Cook, C.N., Nguyen, V.M., Walsh, J.C., Young, N., Cvitanovic, C., Grainger, M.J., Randall, N.P., Muir, M., Kadykalo, A.N., et al., 2023. Environmental evidence in action: on the science and practice of evidence synthesis and evidence-based decision-making. Environ. Evid. 12 (1), 10.

Dai, Y., Cai, X., Zhong, J., MacKenzie, A.R., 2022. Chemistry, street canyon geometry, and emissions effects on NO2 "hotspots" and regulatory "wiggle room". Npj Clim. Atmos. Sci. 5 (1), 102.

Dritsas, E., Trigka, M., Mylonas, P., 2025. Big data analytics in smart cities: Systems, techniques, and applications. In: 2025 IEEE International Smart Cities Conference. ISC2, IEEE, pp. 1–6.

Farhadi, F., Palacin, R., Blythe, P., 2023. Machine learning for transport policy interventions on air quality. IEEE Access 11, 43759–43777.

Feng, Y., Ning, M., Xue, W., Cheng, M., Lei, Y., 2023. Developing China's roadmap for air quality improvement: A review on technology development and future prospects. J. Environ. Sci. 123, 510–521.

Ferraro, P.J., Hanauer, M.M., Miteva, D.A., Nelson, J.L., Pattanayak, S.K., Nolte, C., Sims, K.R., 2015. Estimating the impacts of conservation on ecosystem services and poverty by integrating modeling and evaluation. Proc. Natl. Acad. Sci. 112 (24), 7420–7425.

Freund, M., Zucca, A., Sanson-Fisher, R., Milat, A., Mackenzie, L., Turon, H., 2019. Barriers to the evaluation of evidence-based public health policy. J. Public Health Policy 40 (1), 114–125.
Fu, Z., Yang, X., Ma, Y., Sun, Y., Wang, T., 2025. Integrating explainable AI and causal inference to unveil regional air quality drivers in China. J. Environ. Manag. 390, 126270.
Gu, C., Zhang, L., Xu, Z., Xia, S., Wang, Y., Li, L., Wang, Z., Zhao, Q., Wang, H., Zhao, Y., 2023. High-resolution regional emission inventory contributes to the evaluation of policy effectiveness: A case study in Jiangsu Province, China. Atmos. Chem. Phys. 23 (7), 4247–4269.
Head, B.W., 2016. Toward more "evidence-informed" policy making? Public Adm. Rev. 76 (3), 472–484.
Holnicki, P., Kałuszko, A., Trapp, W., 2016. An urban scale application and validation of the CALPUFF model. Atmos. Pollut. Res. 7 (3), 393–402.
Hong, W.-T., Whyte, J., Xue, J., 2025. A natural language processing–driven framework for policymaking in infrastructure development. J. Constr. Eng. Manag. 151 (5), 04025025.
Hooper, R., Goyal, N., Blok, K., Scholten, L., 2024. A semi-automated approach to policy-relevant evidence synthesis: Combining natural language processing, causal mapping, and graph analytics for public policy. Policy Sci. 57 (4), 875–900.
Hossin, M.A., Du, J., Mu, L., Asante, I.O., 2023. Big data-driven public policy decisions: Transformation toward smart governance. Sage Open 13 (4), 21582440231215123.
Hou, C., Zhang, F., Li, Y., Li, H., Mai, G., Kang, Y., Yao, L., Yu, W., Yao, Y., Gao, S., et al., 2025. Urban sensing in the era of large language models. Innov. 6 (1).
Houghton, L.C., Paniagua-Avila, A., 2023. Why and how epidemiologists should use mixed methods. Epidemiology 34 (2), 175–185.
Hussien, M.M., Melo, A.N., Ballardini, A.L., Maldonado, C.S., Izquierdo, R., Sotelo, M.A., 2025. RAG-based explainable prediction of road users behaviors for automated driving using knowledge graphs and large language models. Expert Syst. Appl. 265, 125914.
Illescas-Martinez, F., Garcia, L., Garcia-Sanchez, A.-J., Asorey-Cacheda, R., Garcia-Haro, J., 2025. Air quality forecasting in non-monitored urban areas through machine and deep-learning model. Expert Syst. Appl. 284, 127749.
Khreis, H., Sanchez, K.A., Foster, M., Burns, J., Nieuwenhuijsen, M.J., Jaikumar, R., Ramani, T., Zietsman, J., 2023. Urban policy interventions to reduce traffic-related emissions and air pollution: A systematic evidence map. Environ. Int. 172, 107805.
Kiekens, A., Dierckx de Casterlé, B., Vandamme, A.-M., 2022. Qualitative systems mapping for complex public health problems: A practical guide. PloS One 17 (2), e0264463.
Kruse, T., Dechezleprêtre, A., Saffar, R., Robert, L., 2022. Measuring environmental policy stringency in OECD countries: An update of the OECD composite EPS indicator. OECD Econ. Dep. Work. Pap. (1703).
Lewis, A.C., Hopkins, J.R., Carslaw, D.C., Hamilton, J.F., Nelson, B.S., Stewart, G., Dernie, J., Passant, N., Murrells, T., 2020. An increasing role for solvent emissions and implications for future measurements of volatile organic compounds. Phil. Trans. R. Soc. A 378 (2183), 20190328.
Li, J., Chen, T., Zhang, H., Jia, Y., Chu, Y., Yan, Y., Zhang, H., Ren, Y., Li, H., Hu, J., et al., 2024. Nonlinear effect of NOx concentration decrease on secondary aerosol formation in the Beijing-Tianjin-Hebei region: Evidence from smog chamber experiments and field observations. Sci. Total Environ. 912, 168333.
Lončarević, Š., Ilinčić, P., Šagi, G., Lulić, Z., 2022. Problems and directions in creating a national non-road mobile machinery emission inventory: A critical review. Sustainability 14 (6), 3471.
Longato, D., Cortinovis, C., Balzan, M., Geneletti, D., 2024. Identifying suitable policy instruments to promote nature-based solutions in urban plans. Cities 154, 105348.
Lu, P., Mishra, S., Xia, T., Qiu, L., Chang, K.-W., Zhu, S.-C., Tafjord, O., Clark, P., Kalyan, A., 2022. Learn to explain: Multimodal reasoning via thought chains for science question answering. Adv. Neural Inf. Process. Syst. 35, 2507–2521.
Mallet, V., Tilloy, A., Poulet, D., Girard, S., Brocheton, F., 2018. Meta-modeling of ADMS-urban by dimension reduction and emulation. Atmos. Environ. 184, 37–46.
Mo, Z., Cui, R., Yuan, B., Cai, H., McDonald, B.C., Li, M., Zheng, J., Shao, M., 2021. A mass balance-based emission inventory of non-methane volatile organic compounds (NMVOCs) for solvent use in China. Atmos. Chem. Phys. Discuss. 2021, 1–26.
Molina-Gómez, N.I., Díaz-Arévalo, J.L., López-Jiménez, P.A., 2021. Air quality and urban sustainable development: The application of machine learning tools. Int. J. Environ. Sci. Technol. 18 (4), 1029–1046.
Nie, Q., Liu, T., 2025. Large language models: Tools for new environmental decision-making. J. Environ. Manag. 375, 124373.
Nowack, P., Konstantinovskiy, L., Gardiner, H., Cant, J., 2021. Machine learning calibration of low-cost NO2 and PM10 sensors: Non-linear algorithms and their impact on site transferability. Atmos. Meas. Tech. 14 (8), 5637–5655.
Orach, K., Duit, A., Schlüter, M., 2020. Sustainable natural resource governance under interest group competition in policy-making. Nat. Hum. Behav. 4 (9), 898–909.
Paolella, D.A., Tessum, C.W., Adams, P.J., Apte, J.S., Chambliss, S., Hill, J., Muller, N.Z., Marshall, J.D., 2018. Effect of model spatial resolution on estimates of fine particulate matter exposure and exposure disparities in the United States. Environ. Sci. Technol. Lett. 5 (7), 436–441.
Pappachan, P., Moslehpour, M., Bansal, R., Rahaman, M., 2024. Transparency and accountability. In: Challenges in Large Language Model Development and AI Ethics. IGI Global, pp. 178–211.
Płonka, M., Kocot, K., Hołda, K., Daniec, K., Nawrat, A., 2025. A comparative evaluation of the effectiveness of document splitters for large language models in legal contexts. Expert Syst. Appl. 272, 126711.
Qiu, M., Weng, Y., Cao, J., Selin, N.E., Karplus, V.J., 2020. Improving evaluation of energy policies with multiple goals: Comparing ex ante and ex post approaches. Environ. Sci. Technol. 54 (24), 15584–15593.
Qu, L., Chai, F., Liu, S., Duan, J., Meng, F., Cheng, M., 2023. Comprehensive evaluation method of urban air quality statistics based on environmental monitoring data and its application. J. Environ. Sci. 123, 500–509.
Ren, R., Ma, J., Zheng, Z., 2025. Large language model for interpreting research policy using adaptive two-stage retrieval augmented fine-tuning method. Expert Syst. Appl. 278, 127330.
Ridzuan, N., Ujang, U., Azri, S., Choon, T.L., 2020. Visualising urban air quality using AERMOD, CALPUFF and CFD models: A critical review. Int. Arch. Photogramm. Remote. Sens. Spat. Inf. Sci. 44, 355–363.
Schindler, S., Mitlin, D., Marvin, S., 2018. National urban policy making and its potential for sustainable urbanism. Curr. Opin. Environ. Sustain. 34, 48–53.
Schlüter, M., Brelsford, C., Ferraro, P.J., Orach, K., Qiu, M., Smith, M.D., 2023. Unraveling complex causal processes that affect sustainability requires more integration between empirical and modeling approaches. Proc. Natl. Acad. Sci. 121 (29), e2215676120.
Schoenegger, P., Tuminauskaite, I., Park, P.S., Bastos, R.V.S., Tetlock, P.E., 2024. Wisdom of the silicon crowd: LLM ensemble prediction capabilities rival human crowd accuracy. Sci. Adv. 10 (45), eadp1528.
Shah, S.B., Thapa, S., Acharya, A., Rauniyar, K., Poudel, S., Jain, S., Masood, A., Naseem, U., 2024. Navigating the web of disinformation and misinformation: Large language models as double-edged swords. IEEE Access.
Shahbazi, H., Ganjiazad, R., Hosseini, V., Hamedi, M., 2017. Investigating the influence of traffic emission reduction plans on tehran air quality using WRF/CAMx modeling tools. Transp. Res. D 57, 484–495.
Shukla, K., Seppanen, C., Naess, B., Chang, C., Cooley, D., Maier, A., Divita, F., Pitiranggon, M., Johnson, S., Ito, K., et al., 2022. ZIP code-level estimation of air quality and health risk due to particulate matter pollution in New York City. Environ. Sci. Technol. 56 (11), 7119–7130.
Sofia, D., Gioiella, F., Lotrecchiano, N., Giuliano, A., 2020. Mitigation strategies for reducing air pollution. Environ. Sci. Pollut. Res. 27 (16), 19226–19235.
Song, C., Liu, B., Cheng, K., Cole, M.A., Dai, Q., Elliott, R.J., Shi, Z., 2023. Attribution of air quality benefits to clean winter heating policies in China: Combining machine learning with causal inference. Environ. Sci. Technol. 57 (46), 17707–17717.
Song, J., Xu, Z., He, M., Feng, J., Shen, B., 2025. Graph retrieval augmented large language models for facial phenotype associated rare genetic disease. NPJ Digit. Med. 8 (1), 543.
Thunis, P., Degraeuwe, B., Pisoni, E., Ferrari, F., Clappier, A., 2016. On the design and assessment of regional air quality plans: The SHERPA approach. J. Environ. Manag. 183, 952–958.
Thunis, P., Degraeuwe, B., Pisoni, E., Trombetti, M., Peduzzi, E., Belis, C., Wilson, J., Clappier, A., Vignati, E., 2018. PM2.5 source allocation in European cities: A SHERPA modelling study. Atmos. Environ. 187, 93–106.
Vámos, C., Scheider, S., Sonnenschein, T., Vermeulen, R., 2024. Ontology of active and passive environmental exposure. Semant. Web 15 (5), 1733–1761.
Volta, M., Giostra, U., Guariso, G., Baldasano, J., Lutz, M., Kerschbaumer, A., Rauterberg-Wulff, A., Ferreira, F., Mendes, L., Monjardino, J., et al., 2022. The greatest air quality experiment ever: Policy suggestions from the COVID-19 lockdown in twelve European cities. PloS One 17 (11), e0277428.
Wang, D.S., Masoud, C.G., Modi, M., Hildebrandt Ruiz, L., 2022. Isoprene–chlorine oxidation in the presence of NOx and implications for urban atmospheric chemistry. Environ. Sci. Technol. 56 (13), 9251–9264.
Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q.V., Zhou, D., et al., 2022. Chain-of-thought prompting elicits reasoning in large language models. Adv. Neural Inf. Process. Syst. 35, 24824–24837.
Wong, A.Y., Eastham, S.D., Monier, E., Selin, N.E., 2025. A geographically weighted Gaussian process regression emulator of the GCHP 13.0.0 global air quality model. EGUsphere 2025, 1–33.
Xie, X., Herrera, M., Xiaoyu, T., James, P., Kassem, M., 2025. Trustworthy urban digital twin: A RAG-based architecture for integrating verifiable knowledge. Available At SSRN 5377584.
Yang, J., Huang, J., Chen, Q., 2024. Construction of time series causal network based on partial rank correlation. Knowl.-Based Syst. 295, 111865.
Yang, C., Kim, Y., 2025. Enhancing topic coherence and diversity in document embeddings using LLMs: A focus on BERTopic. Expert Syst. Appl. 281, 127517.
Zhao, B., Yu, L., Wang, C., Shuai, C., Zhu, J., Qu, S., Taiebat, M., Xu, M., 2021. Urban air pollution mapping using fleet vehicles as mobile monitors and machine learning. Environ. Sci. Technol. 55 (8), 5579–5588.
Zheng, T., Bergin, M.H., Sutaria, R., Tripathi, S.N., Caldow, R., Carlson, D.E., 2019. Gaussian process regression model for dynamically calibrating and surveilling a wireless low-cost particulate matter sensor network in Delhi. Atmos. Meas. Tech. 12 (9), 5161–5181.